

An attenuation-based approach for material basis decomposition in Photon Counting CT

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ABSTRACT

An attenuation-dependent framework for computing a change of material bases in photon-counting computed tomography is presented. This is achieved by minimising the Kullback-Leibler divergence between two distributions parametrised by the material basis decompositions. It is demonstrated that attenuation has a significant effect on the basis transformation. The special case of extracting dual contrast agents I/Gd is investigated in a simulation study with the Gammex phantom and an anthropomorphic phantom.

I. BACKGROUND

Conventional *Computed Tomography* (CT) detectors, so-called *Energy Integrating Detectors* (EIDs), measure the linear attenuation along multiple projection lines through the body, and use this information for tomographic reconstruction of a single 2D-slice [2], with the possibility of stacking slices to obtain full 3D-resolution of the body [10]. Today, a new generation of *Photon Counting Computed Tomography* (PCCT) detectors, capable of measuring the energy of individual X-ray photons, is under development [5, 12]. It has been demonstrated that the additional spectral information provided by PCCT comes with many advantages, such as a higher contrast-to-noise ratio, reduced dose, as well as enhanced spatial resolution [8, 1]. Furthermore, it has been successfully demonstrated that PCCT outperforms conventional EID protocols (such as Dual Energy CT or DECT) at *Material basis Decomposition* (MD) tasks [9]. Here, MD refers

to various techniques of expressing the energy-dependent linear attenuation as a linear combination of two or more material basis functions; this is particularly useful for so-called k-edge imaging, where *Contrast Agents* (CAs) such as Iodine (I) or Gadolinium (Gd) that have a clear spectral signature given by their k-edge, facilitating their detection. The k-edge refers to an abrupt jump in attenuation at a specific energy [7] (at 33.2 and 50.2 keV for iodine and gadolinium, respectively). The ability to separate multiple CAs with k-edge imaging may have various diagnostic applications: most notably in multi-contrast imaging, where two CAs (usually Iodine and Gadolinium) are injected with a delay to study different perfusion phases in an organ. An example of this is biphasic liver imaging, where this technique is used to identify lesions in the liver. Other diagnostic applications of the general 3 MD problem include the identification of calcified plaque in the coronary arteries [7] or the detection of bone-marrow edema associated with fractures [4].

In this paper, we investigate a novel approach to effectively perform a change basis, going from a 2-material MD to one involving three or possibly four basis materials by extracting CAs, thus obtaining new contrast basis images without calibration. This is achieved by identifying regions of interest (ROIs) where it is deemed plausible to have multi-CA presence. Then, using organ specific reference values, the MD can be expressed as a sum of a non-contrast and a contrast contribution. The excess basis maps, i.e. contrast contributions, are

then identified with real CAs. As is shown later, this transformation between excess basis materials and CAs is generally non-linear and attenuation-dependent. For this reason, it is advantageous to solve this problem in the sinogram domain.

II. METHOD

A. Material Basis decompositions

There exist multiple algorithms for performing MD in CT. Most of them, in some way or another, assume that the linear mass attenuation coefficient $\mu(E)$ can be approximated as a linear combination of several basis materials:

$$\mu(x, y, E) = \sum_{j=1}^N a_j(x, y) f_j(E) \quad (1)$$

, typically with $N = 2$ or 3 . Hence, the basis images $a_j(x, y)$ capture the spatial distribution of the j -th material in the MD. In a PCCT-detector with M energy bins, the number of photons detected in bin m ($1 \leq m \leq M$) of a detector element over time τ can, under idealized conditions (no pile-up and no charge-sharing), be modeled with a Poisson distribution [11]. Hence, the probability of detecting a vector of counts $\mathbf{n} = (n_1, \dots, n_M)$ is

$$p(\mathbf{n}; \boldsymbol{\lambda}) = \prod_{i=1}^M p(n_i; \lambda_i) = \prod_{i=1}^M \frac{\lambda_i^{n_i} e^{-\lambda_i}}{(n_i)!}. \quad (2)$$

In a PCCT-detector, the parameters λ_i are determined by the initial spectrum, material path-lengths and the energy-dependent response functions for each energy bin. This is described by the Beer-Lambert law:

$$\begin{aligned} \lambda_i = \lambda_i(\mathbf{A}) &= \int \omega_i(E) e^{-\sum_{j=1}^N A_j f_j(E)} dE, \\ A_i(\ell) &= \int_{\ell} a_i(x, y) d\ell. \end{aligned} \quad (3)$$

Here, $\omega_i(E)$ is the combined effect of the source and the detector, N the number of materials, A_j the path-lengths, and $f_j(E)$ the normalized basis

functions. There exist multiple approaches for estimating the basis images from data [3], and they generally require calibration.

B. Theoretical framework

From eq. (3), it is evident that distinct MDs in general give rise to different $\boldsymbol{\lambda}$ -vectors and thus different probability distributions of counts. Intuitively, a successful transformation between material bases aims to minimize this discrepancy. Hence, two different probability distributions are introduced: $P(\mathbf{n}; \boldsymbol{\lambda})$ corresponding to the preexisting 2-material MD $\mathbf{a}$ and a model distribution $Q(\mathbf{n}; \tilde{\boldsymbol{\lambda}})$ which is parameterized by the CA-dependent target MD $\tilde{\mathbf{a}}$. It is further assumed that $\mathbf{a}$ can be decomposed into the contributions of contrast ($\Delta\mathbf{a}$) and non-contrast materials ($\mathbf{a}^0$) as follows:

$$\mathbf{a} = \mathbf{a}^0 + \Delta\mathbf{a} = (a_1^0 + \Delta a_1, a_2^0 + \Delta a_2). \quad (4)$$

This can be achieved by using already known organ-specific estimates for $\mathbf{a}^0$, possibly in combination with image segmentation and other machine learning methods. Alternatively, one could use consecutive scans before and after injection to estimate the contrast contribution. Additionally, let $\mathbf{a}^C = (a_1^C, a_2^C)$ denote contrast agents in the target MD and let

$$\tilde{\mathbf{a}} = (\mathbf{a}^0, 0, 0) + (0, 0, \mathbf{a}^C) = (a_1^0, a_2^0, a_1^C, a_2^C). \quad (5)$$

That is, while fixing the non-contrast contributions $\mathbf{a}^0$ in $\tilde{\mathbf{a}}$, the contrast contributions $\Delta\mathbf{a}$ are transformed to de facto contrast maps $\mathbf{a}^C$. However, since this problem is attenuation-dependent, the images $\mathbf{a}^0$ and $\Delta\mathbf{a}$ are forward projected with the Radon transform, yielding background sinograms $\mathbf{A}^0 = \mathcal{R}[\mathbf{a}^0]$ and excess basis sinograms $\Delta\mathbf{A} = \mathcal{R}[\Delta\mathbf{a}]$. Hence, we want to find the mapping:

$$\begin{aligned} f : \mathbb{R}^4 &\rightarrow \mathbb{R}^2 \\ (\mathbf{A}^0, \Delta\mathbf{A}) &\mapsto \mathbf{A}^C. \end{aligned} \quad (6)$$

for which the related distributions $\mathcal{P}(\mathbf{n}; \boldsymbol{\lambda})$ and $\mathcal{Q}(\mathbf{n}; \tilde{\boldsymbol{\lambda}})$ are as similar as possible by some reasonable similarity score. The mapping f can be estimated from data, but also from analytical methods. For fixed $(\mathbf{A}^0, \Delta \mathbf{A})$, the ground truth distribution $\mathcal{P}(\mathbf{n}; \boldsymbol{\lambda})$ is determined, whereas $\mathcal{Q}(\mathbf{n}; \tilde{\boldsymbol{\lambda}})$ is parametrized by CAs $\mathbf{A}^C$. For brevity, let $\boldsymbol{\lambda} := \boldsymbol{\lambda}(\mathbf{A}^0, \Delta \mathbf{A})$ and $\tilde{\boldsymbol{\lambda}}(\mathbf{A}^C) := \tilde{\boldsymbol{\lambda}}(\mathbf{A}^0, \mathbf{A}^C)$. Then the prediction $\hat{\mathbf{A}}^C$ is chosen as the minimizer of the following objective:

$$\begin{aligned} \hat{\mathbf{A}}^C &= f(\mathbf{A}^0, \Delta \mathbf{A}) \\ &:= \underset{\mathbf{A}^C}{\operatorname{argmin}} \mathcal{D}_{sym} \left[\mathcal{P}(\mathbf{n}; \boldsymbol{\lambda}) \parallel \mathcal{Q}(\mathbf{n}; \boldsymbol{\lambda}(\mathbf{A}^C)) \right]. \end{aligned} \quad (7)$$

Where $\mathcal{D}_{sym}$ is defined as the symmetrized Kullback-Leibler divergence

$$\begin{aligned} \mathcal{D}_{sym}(\mathcal{P}, \mathcal{Q}) &:= \frac{1}{2} \left(\mathcal{D}_{KL}(\mathcal{P} \parallel \mathcal{Q}) + \mathcal{D}_{KL}(\mathcal{Q} \parallel \mathcal{P}) \right) \\ , \text{ and } \mathcal{D}_{KL}(\mathcal{P} \parallel \mathcal{Q}) &= \mathbb{E}_{x \sim \mathcal{P}} \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} \end{aligned} \quad (8)$$

This is a reasonable objective since the KL-divergence of $\mathcal{P}$ from $\mathcal{Q}$ is a positive measure of how much the distributions $\mathcal{P}$ and $\mathcal{Q}$ differ and is zero if and only if $\mathcal{P}$ and $\mathcal{Q}$ are identical. By using a symmetrized version, $\mathcal{P}$ and $\mathcal{Q}$ are valued equally. We present a Newton-based minimization scheme. Thus, the gradient and Hessian of $\mathcal{D}_{sym}$ with respect to $\mathbf{A}^C$ must be calculated. From equation (2), it follows that

$$\mathcal{D}_{sym}(\mathcal{P}, \mathcal{Q}) = \frac{1}{2} \sum_{i=1}^M \left(\lambda_i \log \frac{\lambda_i}{\tilde{\lambda}_i} - \tilde{\lambda}_i \log \frac{\tilde{\lambda}_i}{\lambda_i} \right) \quad (9)$$

By differentiating (9) and using the chain rule, the gradient $\nabla_{\mathbf{A}} \mathcal{D}_{sym}$ and Hessian $\mathcal{H}_{\mathbf{A}}(\mathcal{D}_{sym})$ can be expressed in terms of the quantities $(\mathcal{J}_{ij}) = \partial_i \tilde{\lambda}_j$ and $(\mathcal{H}_{ijk}) = \partial_i \partial_j \tilde{\lambda}_k$ as:

$$\partial_i \mathcal{D}_{sym} = \frac{1}{2} \sum_{j=1}^M \alpha_j \mathcal{J}_{ij} \quad (10)$$

$$\partial_i \partial_j \mathcal{D}_{sym} = \frac{1}{2} \sum_{k=1}^M \left(\alpha_k \mathcal{H}_{ijk} + \beta_k \mathcal{J}_{ik} \mathcal{J}_{jk} \right)$$

, where $\alpha_j = 1 - \frac{\lambda_j}{\tilde{\lambda}_j} + \log \frac{\tilde{\lambda}_j}{\lambda_j}$, $\beta_j = \frac{\lambda_j + \tilde{\lambda}_j}{\tilde{\lambda}_j^2}$, and

$$\begin{aligned} \mathcal{J}_{ij} &= - \int \omega_j f_i e^{-\sum_{m=1}^N A_m f_m} dE \\ \mathcal{H}_{ijk} &= \int \omega_k f_i f_j e^{-\sum_{m=1}^N A_m f_m} dE. \end{aligned} \quad (11)$$

Using the gradient and Hessians from (10), we then use Newton's method to estimate the minimizer using the update scheme:

$$\mathbf{A}^{n+1} = \mathbf{A}^n - \alpha [\mathcal{H}_{\mathbf{A}}(\mathcal{D}_{sym})]^{-1} (\nabla_{\mathbf{A}} \mathcal{D}_{sym}) \quad (12)$$

where $\alpha \in (0, 1)$ either is a fixed hyperparameter or can be determined using the Armijo-Goldstein condition.

C. Simulated phantoms

A simulation study was performed. The small Gammex-phantom and an anthropomorphic phantom were simulated with CatSim, a software for CT simulations developed by GE Healthcare. Analysis was restricted to the case of extracting dual CAs *Iodine* (I) and *Gadolinium* (Gd) from images expressed in the material bases *Polyethylene* (PE) and *Polyvinyl chloride* (PVC). First, ground truth basis maps with water (Wa) and calcium (Ca), as well as CA maps with anatomically relevant concentrations of I and Gd, were produced using the segmentation software Total Segmentator [6]. The small Gammex phantom (diameter = 200 mm) was simulated with 8 inserts (diameter = 2 cm) containing concentrations of up to 10 mgI/ml and 4 mgGd/ml, see fig. 1. In the anthropomorphic phantom, concentrations up to 10 mgI/ml and 1 mgGd/ml were added to specified organs (e.g., kidneys, blood) using organ

segmentation. The PCCT chain, as well as an MD into PE/PVC $\mathbf{a}$ maps, was then simulated in Cat-Sim. In both cases, perfect non-contrast images $\mathbf{a}^0$ were obtained by performing the same steps above, using only the ground-truth Wa/Ca maps. Then the contrast contributions were calculated using the relationship $\Delta \mathbf{a} = \mathbf{a} - \mathbf{a}^0$.

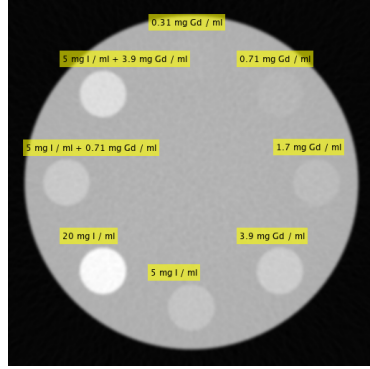


Fig. 1: Concentrations of I/Gd samples in the small gammex phantom used for the simulation study.

III. RESULTS

A. Theoretical results

As in the simulation study, theoretical analysis was restricted to the case of separating I/Gd from PE/PVC maps. In fig. 2A, the log-KL divergence is plotted and its minimum highlighted for fixed $\mathbf{A}^0 = (100, 10)$ (mm) and $\Delta \mathbf{A} = (10, 10)$ (mm). In fig. 2B, the attenuation curve for the contrast contribution $\Delta \mathbf{A}$ expressed in the PE/PVC basis is compared to that of the predicted contrast $\hat{\mathbf{A}}^C$. It is observed that the curves align well for energies above the k-edge of I at $E \approx 30$ keV; however, for $E < 30$ keV, they diverge. That said, this part of the spectrum is highly attenuated with a transmission of $\sim 10^{-5}$ through 30 cm of water at 30 keV compared to $\sim 10^{-3}$ at 50 keV, and hence has a limited effect on the model. Figures 4 and 3 visualize relationships between CAs $\hat{\mathbf{A}}^C$ and $\mathbf{A}^0$ as well as $\Delta \mathbf{A}$. Around the origin, a roughly linear relationship was observed between $\hat{\mathbf{A}}^C$ and $\Delta \mathbf{A}$.

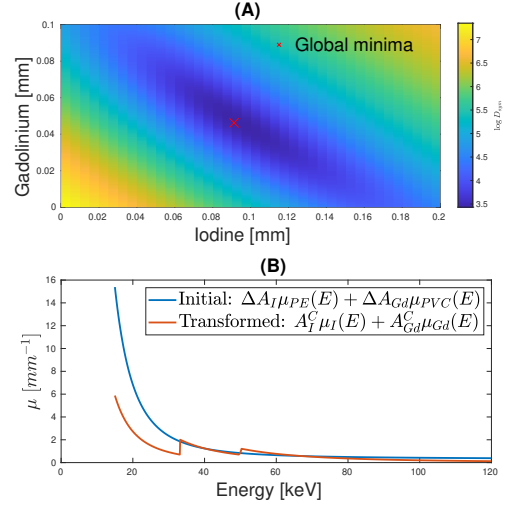


Fig. 2: (A): $\log \mathcal{D}_{sym}$ for fixed $(A_{PE}^0, A_{PVC}^0) = (100, 10)$ (mm) and $(\Delta A_{PE}, \Delta A_{PVC}) = (10, 10)$ (mm), and its minimum at $(\hat{A}_I^C, \hat{A}_{Gd}^C) = (0.083, 0.051)$ (mm). (B): Comparisons of initial (Blue) and transformed (Red) attenuation curves.

The quality of the linear approximation appears to depend on attenuation, and comparing Figures 3A and 3D suggests that the non-linear behavior increases with attenuation. Additionally, attenuation correlates positively with Gd and negatively with I. In fig. 4, the effect of attenuation is shown. In fig. 4A, it can be seen that $\Delta \mathbf{A} = (0, 0)$ maps to zero no matter the background $\mathbf{A}^0$, which is expected. Perhaps more interesting is that for non-zero $\Delta \mathbf{A}$, the attenuation dependence is substantial, and seemingly increasing with $\Delta \mathbf{A}$. Curiously, fig. 4 suggests that for fixed $\Delta \mathbf{A}$, the solution $\hat{\mathbf{A}}^C$ moves along a 1-dimensional curve parameterised by some effective attenuation. Just as in fig. 3, it is observed that Gd increases with attenuation, whereas I decreases.

B. Simulations study

The results for the small Gammex phantom can be seen in figures 5-8. In the final prediction, i.e. fig. 8, I could be extracted successfully. The predicted Gd map has some errors; that said, it is

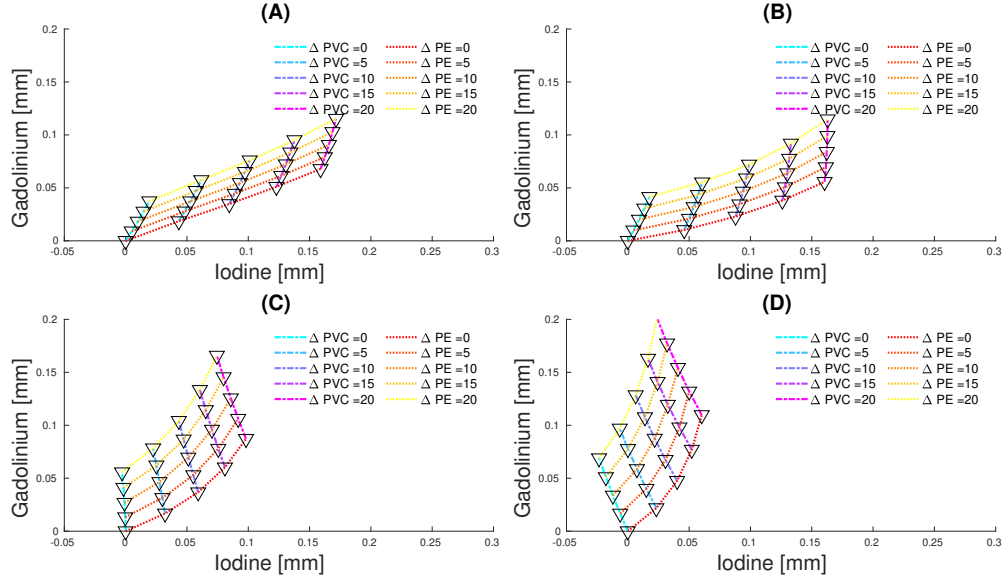


Fig. 3: The effect of linearly changing the contrast contributions $\Delta \mathbf{A} = (\Delta A_{PE}, \Delta A_{PVC})$ on the transformation f at discrete points $\mathbf{A}^0 = (A_{PE}^0, A_{PVC}^0)$. (A): $\mathbf{A}^0 = (0, 0)$, (B): $\mathbf{A}^0 = (200, 0)$, (C): $\mathbf{A}^0 = (0, 50)$, and (D): $\mathbf{A}^0 = (200, 50)$.

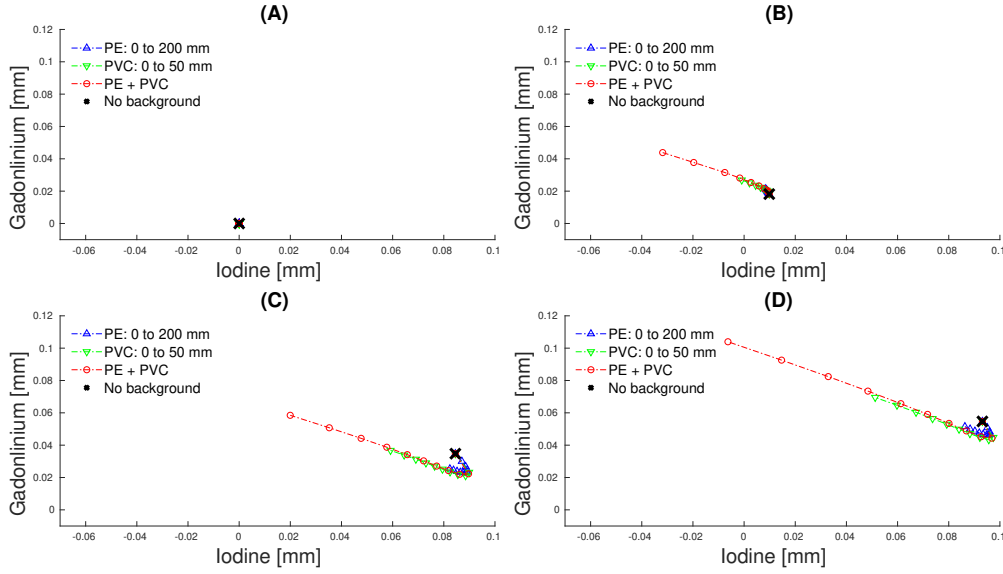


Fig. 4: The effect of linearly changing the background $\mathbf{A}^0 = (A_{PE}^0, A_{PVC}^0)$ on the transformation f at discrete points $\Delta \mathbf{A} = (\Delta A_{PE}, \Delta A_{PVC})$. (A): $\Delta \mathbf{A} = (0, 0)$, (B): $\Delta \mathbf{A} = (10, 0)$, (C): $\Delta \mathbf{A} = (0, 10)$, and (D): $\Delta \mathbf{A} = (10, 10)$.

hard to judge whether these errors are systematic or due to noise. Nevertheless, the anthropomorphic phantom (figures 9-12) proved more challenging. As seen in fig. 12 neither I nor Gd are successfully extracted.

IV. DISCUSSION

An attenuation-dependent method for performing a change of material bases in PCCT has been analysed. It has been demonstrated that attenuation introduces nonlinearities and can have a substantial effect on the basis transformation. It has thus been

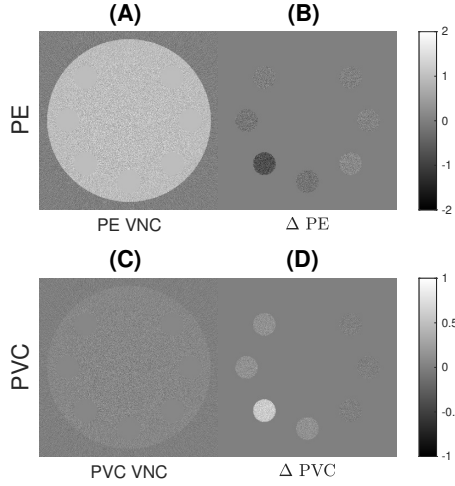


Fig. 5: The non-contrast $\mathbf{a}^0 = (a_{PE}^0, a_{PVC}^0)$ (left) and contrast $\Delta \mathbf{a} = (\Delta a_{PE}, \Delta a_{PVC})$ (right) contributions for the small Gammex phantom.

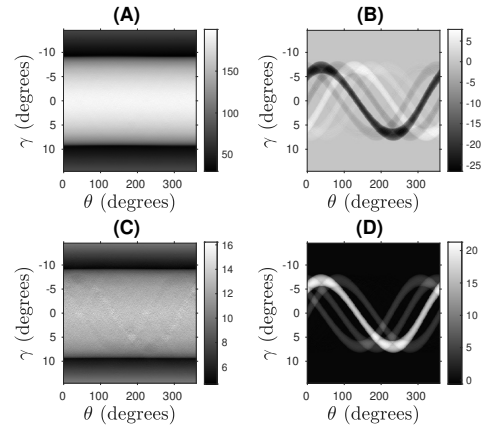


Fig. 6: The corresponding non-contrast $\mathbf{A}^0 = (A_{PE}^0, A_{PVC}^0)$ (left) and contrast $\Delta \mathbf{A} = (\Delta A_{PE}, \Delta A_{PVC})$ (right) sinograms for the small Gammex phantom.

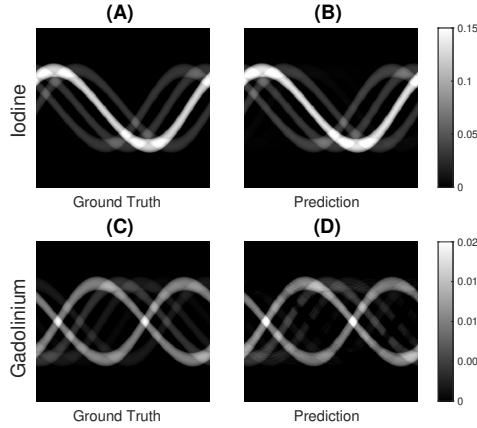


Fig. 7: The initial $\mathbf{A}^C = (A_I^C, A_{Gd}^C)$ (left) and predicted $\hat{\mathbf{A}}^C = (\hat{A}_I^C, \hat{A}_{Gd}^C)$ (right) contrast sinograms for the small Gammex phantom.

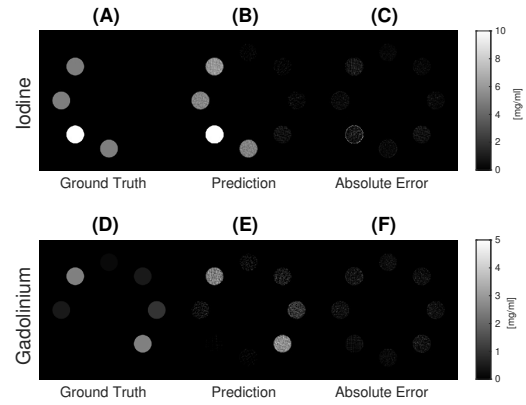


Fig. 8: The initial $\mathbf{a}^C = (a_I^C, a_{Gd}^C)$ (left), predicted $\hat{\mathbf{a}}^C = (\hat{a}_I^C, \hat{a}_{Gd}^C)$ (middle) contrast maps (mg/ml) and the absolute errors (right) for the small Gammex phantom.

An attenuation-based approach for material basis decomposition

shown that a naive, linear approach might fall short in this setting. Even though the attenuation dependence is non-linear, it could be described well with a single effective attenuation. This suggests that a dimensionality reduction is possible, treating only a single attenuation dimension and two material

dimensions.

The attenuation-based approach worked well for the small Gammex phantom. This presents a case of medium attenuation of approximately 20 cm of water. However, for the more complex and attenu-

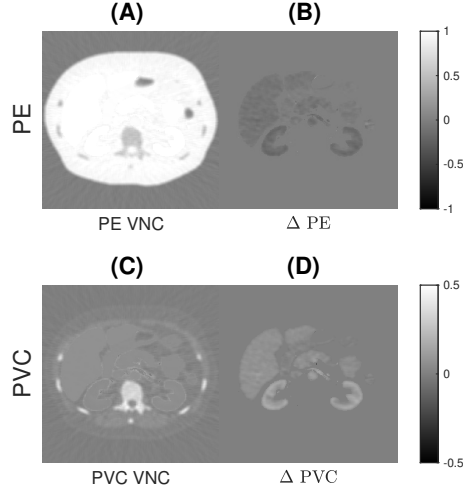


Fig. 9: The non-contrast $\mathbf{a}^0 = (a_{PE}^0, a_{PVC}^0)$ (left) and contrast $\Delta \mathbf{a} = (\Delta a_{PE}, \Delta a_{PVC})$ (right) contributions for the anthropomorphic phantom.

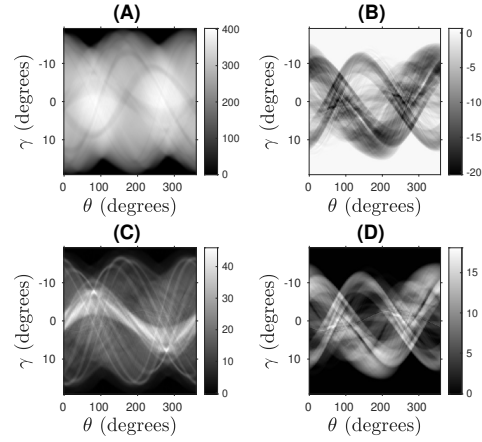


Fig. 10: The corresponding non-contrast $\mathbf{A}^0 = (A_{PE}^0, A_{PVC}^0)$ (left) and contrast $\Delta \mathbf{A} = (\Delta A_{PE}, \Delta A_{PVC})$ (right) sinograms for the anthropomorphic phantom.

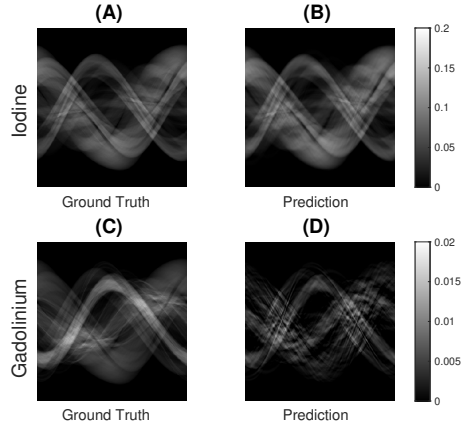


Fig. 11: The initial $\mathbf{A}^C = (A_I^C, A_{Gd}^C)$ (left) and predicted $\hat{\mathbf{A}}^C = (\hat{A}_I^C, \hat{A}_{Gd}^C)$ (right) contrast sinograms for the anthropomorphic phantom.

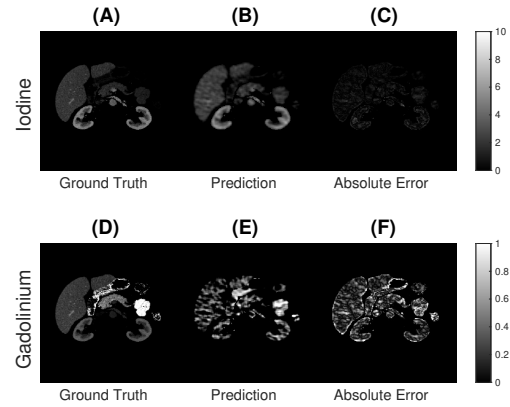


Fig. 12: The initial $\mathbf{a}^C = (a_I^C, a_{Gd}^C)$ (left), predicted $\hat{\mathbf{a}}^C = (\hat{a}_I^C, \hat{a}_{Gd}^C)$ (middle) contrast maps (mg/ml) and the absolute errors (right) for the anthropomorphic phantom.

ating anthropomorphic phantom, it was inaccurate. This suggests a mismatch between the forward model used in the simulation and the attenuation-based approach presented. One possibility is that the simulation does not fully account for the effects of attenuation. However, it could also be the case that this method is over-simplified (no pile-up or charge-sharing) or simply incorrect (maybe one should not minimize the KL-divergence, but use some other objective instead). This highlights a problem with this approach of material basis decomposition: it requires a compatible forward model to function properly.

Furthermore, projecting onto the PE/PVC basis may result in a loss of spectral information since PE/PVC lacks the k-edges present in I/Gd. Therefore, it might be interesting to develop a similar method based on the raw counts instead, avoiding the problem of spectral information loss altogether. However, this poses new challenges. It becomes substantially harder (if not impossible) to extract contrast and non-contrast information from pure counts. Another, perhaps more feasible option is to combine these two approaches, using both PE/PVC maps as well as the raw counts to allow for contrast/non-contrast separation while avoiding loss of spectral information.

V. CONCLUSION

An attenuation-based approach for material basis decomposition for photon counting CT has been presented. It was shown that the effect of attenuation on the basis transformation is complicated and highly non-linear, and that, therefore, a naive linear mapping between bases might fail in the regime of high attenuation. A simulation study was conducted where the method worked well for medium attenuation, but did poorly in regimes of high attenuation. It was speculated that the effectiveness of the model might depend heavily on the spectral properties of the materials considered. It was furthermore noted that it might be interesting to consider attenuation-dependent MD at an earlier stage in the chain using raw counts, possibly accompanied by the PE/PVC maps.

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VII. APPENDIX A - DERIVING EQUATION 9

Assume that $P(\mathbf{n}; \lambda)$ and $Q(\mathbf{n}; \tilde{\lambda})$ be multivariate Poisson distributions. The result is a straightforward computation:

$$\begin{aligned} D_{KL}(P||Q) &= \sum_{\mathbf{n} \in \mathbb{R}^M} P(\mathbf{n}) \log \frac{P(\mathbf{n})}{Q(\mathbf{n})} \\ &= \mathbb{E}_P \left[\log P(\mathbf{n}) - \log Q(\mathbf{n}) \right] \\ &= \mathbb{E}_P \left[\sum_{i=1}^M n_i \log(\lambda_i) - \lambda_i - (n_i \log(\tilde{\lambda}_i) - \tilde{\lambda}_i) \right] \\ &= \sum_{i=1}^M \mathbb{E}_P \left[n_i \log\left(\frac{\lambda_i}{\tilde{\lambda}_i}\right) - (\lambda_i - \tilde{\lambda}_i) \right] \\ &= \sum_{i=1}^M \lambda_i \log\left(\frac{\lambda_i}{\tilde{\lambda}_i}\right) - (\lambda_i - \tilde{\lambda}_i) \end{aligned}$$

$$\begin{aligned} \lambda_i &= \lambda_i(\mathbf{A}^0, \tilde{\mathbf{A}}) \\ &= \int_{\mathbb{R}} \omega_i(E) e^{-\sum_{j=1}^N A_j f_j(E)} dE \\ &= \int_{\mathbb{R}} \underbrace{\omega_i(E) e^{-A_1^0 f_1(E) - A_2^0 f_2(E)}}_{:= \hat{\omega}_i(E)} e^{-\sum_{j=3}^N A_j f_j(E)} dE \\ &= \int_{\mathbb{R}} \hat{\omega}_i(E) e^{-\sum_{j=3}^N A_j f_j(E)} dE \end{aligned}$$

Hence, if $N = 4$, this approach reduces the search space from 4 dimensions to 2 dimensions, which may make this problem more feasible.

A. Appendix C - results using Wa/I basis instead

VIII. APPENDIX B - A METHOD BASED ON RAW COUNTS

Given a vector of raw counts $\mathbf{n}$, the values $\mathbf{A} = (\mathbf{A}^0, \tilde{\mathbf{A}})$ can be estimated using maximum likelihood. The log likelihood function is given by

$$\mathcal{L}(\mathbf{n}; \lambda) = C + \sum_{i=1}^M n_i \log(\lambda_i) - \lambda_i$$

where C is a constant, and λ_i are given by 3. Thus, the objective is

$$(\mathbf{A}^0, \tilde{\mathbf{A}}) = \operatorname{argmax}(\mathcal{L}(\mathbf{n}; \lambda(\mathbf{A}))).$$

This optimisation problem can be difficult to solve. However, if complementary information is provided, it might become more feasible. For example, if $\mathbf{A}^0$ is known, then the expressions for λ_i are simplified:

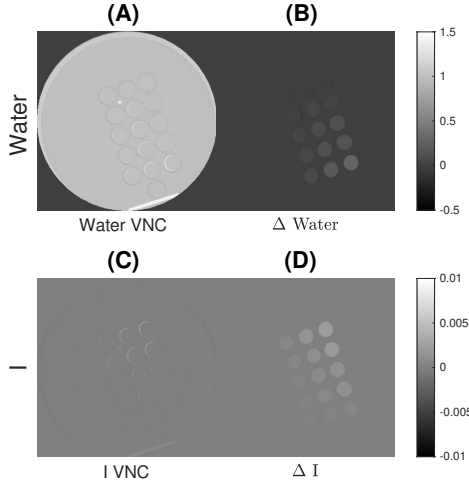


Fig. 13: The non-contrast $\mathbf{a}^0 = (a_{Wa}^0, a_I^0)$ (left) and contrast $\Delta \mathbf{a} = (\Delta a_{Wa}, \Delta a_I)$ (right) contributions for the UZB phantom.

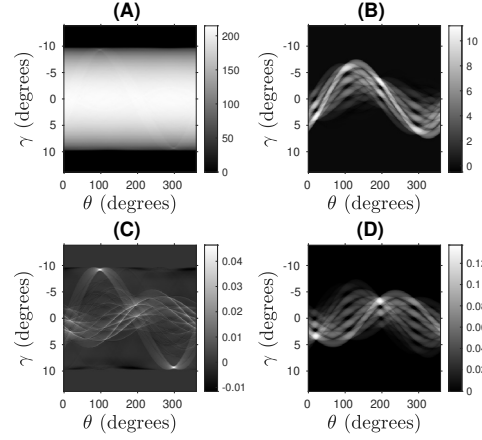


Fig. 14: The corresponding non-contrast $\mathbf{A}^0 = (A_{Wa}^0, A_I^0)$ (left) and contrast $\Delta \mathbf{A} = (\Delta A_{Wa}, \Delta A_I)$ (right) sinograms.

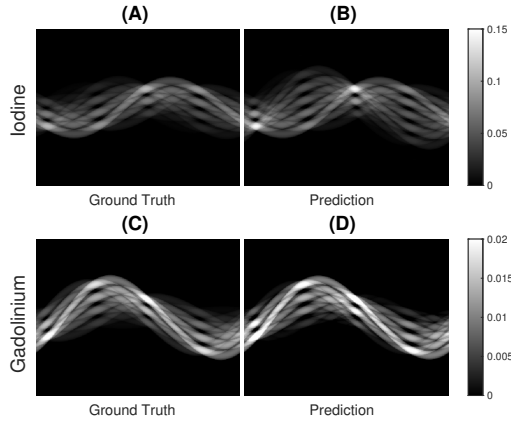


Fig. 15: The initial $\mathbf{A}^C = (A_I^C, A_{Gd}^C)$ (left) and predicted $\hat{\mathbf{A}}^C = (\hat{A}_I^C, \hat{A}_{Gd}^C)$ (right) contrast sinograms.

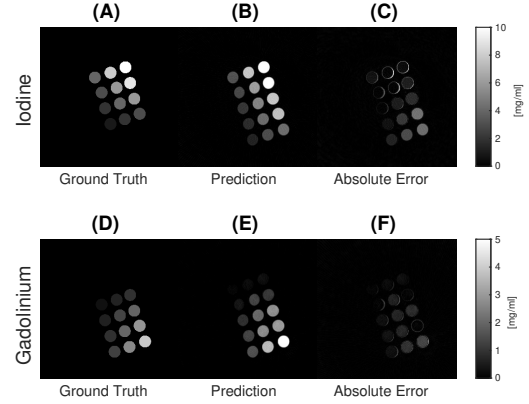


Fig. 16: The initial $\mathbf{a}^C = (a_I^C, a_{Gd}^C)$ (left), predicted $\hat{\mathbf{a}}^C = (\hat{a}_I^C, \hat{a}_{Gd}^C)$ (middle) contrast maps and the absolute errors (right).